Predicting Cherry Quality Using Siamese Networks

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***Abstract*—The cherry industry is a rapidly growing sector of New Zealand's export merchandise and, as such, the accuracy with which pack-houses can grade cherries during processing is becoming increasingly critical. Conventional computer vision systems are usually employed in this process, yet they fall short in many respects, still requiring humans to manually verify the grading. In this work, we investigate the use of deep learning to improve upon the traditional approach. The nature of the industry means that the grade standards are influenced by a range of factors and can change on a daily basis. This makes conventional classification approaches infeasible (as there are no fixed classes) so we construct a model to overcome this. We convert the problem from classification to regression, using a Siamese network trained with pairwise comparison labels. We extract the model embedded within to predict continuous quality values for the fruit. Our model is able to predict which of two similar quality fruit is better with over 88% accuracy, only 5% below the self-agreement of a human expert.**

I. Introduction

New Zealand's horticulture industry exports reached 6.2 billion dollars in 2019, and account for more than 10% of the country's total merchandise export income. The cherry sector is a relatively new addition to New Zealand's exports, yet is rapidly growing, with export income increasing from 27 to 69 million dollars over the last ten years [1]. As such, while the demand for high quality cherries has dramatically increased, the technology backing production has lagged behind. In this work, we consider the grading of cherries within a pack-house – which is to say, we are concerned with estimating the quality of fruit that pickers have already harvested.

Current cherry grading systems use a variety of techniques but, over the last two decades, pack-houses have incorporated computer vision into the process. Complex and expensive hardware (in this work we use the *RP360* grader [2]) aims to rotate the cherries, capturing multiple images in order to see their entire surface. The stems make this a considerable challenge that no one has entirely solved. See Figure 1 for sample images of some typical cherries. No existing system performs well enough to eliminate the need for manual grading of the fruit. In comparison to the development efforts put into the related hardware, software is generally quite rudimentary, and there is strong potential for more robust and flexible methods to provide serious improvement.

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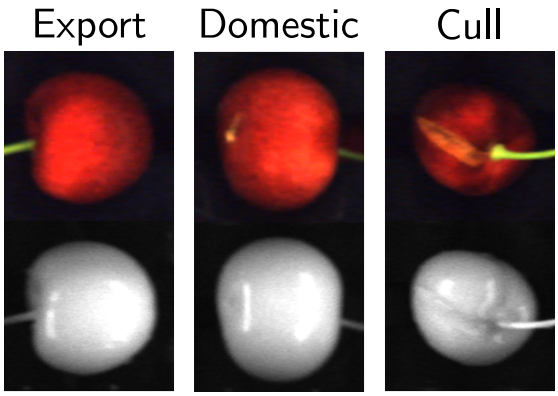
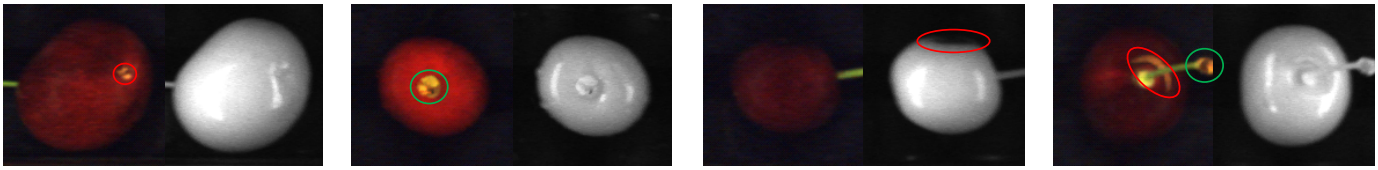


Fig. 1. Typical cherries for the three grades, as captured by the RP360 grader. Top image is RGB, bottom is near infrared (NIR).

In the last decade, deep learning (DL) has dominated image classification (along with many other areas of computer vision). This motivates its use as a means of improving the quality of cherry grading beyond what the conventional approaches can offer. However, normal image classification methods are infeasible here, as the grade standards – the quality thresholds for a fruit to be exported, sold domestically, or culled – can change on a daily basis. This can be due to a range of factors, such as the overall quality of the season, standards of different buyers, daily fluctuation in product demand (both export and domestic), and more. As such, tackling this problem (of flexible classes) is the main focus of our work. To do so, we transform the problem from classification into regression, designing a model that predicts continuous quality values. The issue with such an approach is that it is very difficult (or even impossible) for a human to label fruit on a consistent, continuous scale. Therefore, we train our model using pairwise comparisons – which are much easier for a human to provide. We develop a Siamese network that is trained using these comparisons, after which the base model inside the network can be used to predict quality values. We find that, although the model does not yet reach human performance, it comes close.

II. Fruit Grading

The existing system our model must work with consists of hardware (the RP360) that captures both colour and near



(a) Small nose split (b) Deceptive stem (c) Missed rot (d) Ignored crack

Fig. 2. Common mistakes made by the existing grading software (but by no means a comprehensive list). Red outlines indicate true defects, while green ones highlight potential false positives. (a) A small split in the nose of the cherry that would likely be overlooked. (b) A perfect cherry seen as defective, due to the image being taken such that the end of the stem appears to be a defect. (c) The rot visible in the NIR image blends into the background, and so would be ignored. (d) A more complex issue. The existing system must be setup to ignore "brown" that is touching "green", so that the end of the stem is not thought to be a defect. Unfortunately, this also means the crack behind the stem is ignored.

infrared (NIR) images of the cherry (see Figure 1), as well as software that is used to set thresholds. The current software is fairly limited, extracting features such as colour, circularity, and the amount of brown on the fruit. The colour values corresponding to "brown" must be painstakingly set and adjusted many times throughout the day by an operator in order to deal with the varied colour of cherries and fluctuating lighting conditions (refer to the differences visible in Figure 2, and later Figure 4), with the added difficulty that brown is very close to the red skin colour of a cherry. New Zealand's fruit exporting standards require pack-houses to grade with no less than 90% accuracy, a feat that is not achievable using the current system without the aid of staff manually verifying the grading. To better illustrate why the existing system is unable to meet these performance requirements, consider the examples displayed in Figure 2.

A recent review of fruit quality grading systems that use machine learning [3] identified several main approaches of examining fruit: hyperspectral imaging, multispectral imaging, ultrasound, magnetic resonance imaging, electric tomography, or computed tomography. The baseline is multispectral imaging, which captures a fixed number of bands of the electromagnetic spectrum (for example, red, green, blue, and possibly NIR), and it is what we examine in this work, as the RP360 provides colour and NIR images. Multispectral systems are fast, affordable, and provide good information about the external quality of the fruit. Using NIR, it is also possible to detect rot that is close to the surface (which is almost always where it is located in cherries). The other approaches can provide additional information, typically about the internal quality of the fruit [4]–[6] or the maturity [7] (which, for cherries, can already be estimated using NIR). However, these approaches are still infeasible for commercial use, due to factors such as the affordability of equipment, health and safety, being too slow for use in real-time systems, overall reliability, and imaging quality [8]–[11].

A variety of tasks broadly related to our work have been explored, such as recognising different types of fruit [12] and analysing whether fruit are ready to be harvested [13]. We, however, focus on quality evaluation, which can be based on a number of factors, such as: colour, morphological features (i.e., size, shape), texture, and/or defect detection. Colour, size, and shape are easily handled without the use of DL, and so our work concerns itself with defect detection.

There is some academic work concerning the use of machine learning for cherries. A two-layer feed forward neural network (based on hyperspectral transmission images) was used to determine whether cherries have pits [14]; a Bayesian classifier has been used to detect branches of cherry trees for harvesting [15]; and disease has been predicted from environment conditions [16]. Beyond these, work involving cherries is increasingly unrelated to grading.

Considering DL, out of the 34 machine learning approaches used for fruit quality evaluation found in a recent review [3], 16 had a neural network based approach. However, only one utilised a convolutional neural network (CNN). It appears that the overall field of fruit image processing is behind the current trends (of using DL), though it is now beginning to catch up [17]. Even so, in terms of academic DL work, researchers have done little involving cherries. A CNN has been used to classify the shape of cherries [18] and, while this is a simple task, an overall accuracy of 99.4% was achieved. Additionally, a CNN that examines cherry crops has been used to estimate harvest yields [19]. As far as we are aware, this is the extent that DL has been applied to cherries – with no serious attempts for its use in quality evaluation, indicating a serious gap in the field.

III. Siamese Networks for Quality Ranking

The main focus of this work is to deal with the flexible grade standards present in the cherry industry. Siamese networks are a major component in our process. First introduced for signature verification [21], Siamese networks consist of two identical sub models (usually neural networks) that are joined together in such a way that their outputs are compared against each other (see Figure 3), providing a similarity score. Our use of Siamese networks is inspired by ranking tasks for information retrieval (e.g., search engines), where the model learns to predict which of two (or more) provided inputs is "better". The first major use of a Siamese neural network for this kind of task was RankNet [22]. More recently, Siamese networks with CNNs as their base model have seen success in object position tracking [23] and the ranking of image quality [24]. The latter is most relevant to us, except instead of ranking based on the quality of the image, we rank based on the quality of the cherry it contains. We train the network using pairwise comparisons and then extract the embedded

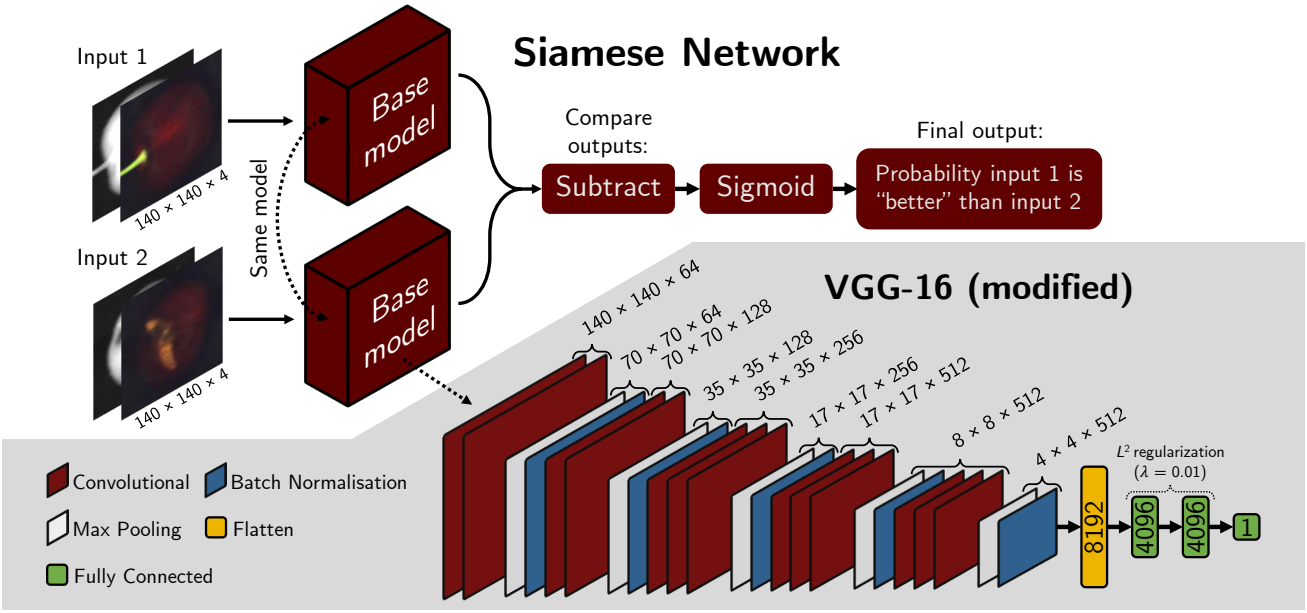


Fig. 3. Network architecture. The top section outlines a Siamese network, while the bottom describes the base model we used (based on the VGG-16 [20] architecture, slightly modified to match our input image shape and to include batch normalisation layers). Note that the exact same network (with the same learned parameters) is used for the base model in both branches of the Siamese network.

model to produce a single scalar that represents quality, which converts the problem from classification (of three fixed classes) to regression (predicting a continuous quality score for each fruit). This mitigates the problem of flexible grade standards by allowing us to set and adjust the class thresholds (in terms of predicted quality values) after the model has been trained.

A. Data

We collected images during the 2019-2020 cherry season and performed most labelling on-site. The images are 140 by 140 pixels, with the cherry centred, and have 4 colour channels (RGB + NIR). Fruit are labelled with an overall quality class (export, domestic, or cull), and with any individual defects visible, though we do not use this directly in our work. We manually double-checked these labels during pre-processing. Note that the labelled quality classes in no way account for flexible grade standards, as the labelling only corresponds to the grades at a single point in time. As the quality standards regularly change, these labels quickly become inaccurate. For instance, the export class here only contains cherries without any defects, while in reality there are times when fruit with very minor defects can still be exported. In total, there were 2205 export, 1109 domestic, and 1406 cull cherries. We split the labelled data into 3801 training, 471 validation, and 502 testing images, each set containing roughly the same distribution of grades and different types of defects.

To generate pairwise comparisons between fruit, we built a simple program that presents the user with two randomly sampled images from within the *same* labelled quality class (to get comparisons for fruit from *different* labelled quality classes, all we need to do is compare their two labels). The user must then decide whether the first cherry is better or worse than the second. Additionally, they can also select that the two are (roughly) equal – for instance, when neither of the fruit are objectively better than the other. The process is set up such that all images receive roughly the same number of comparisons. We then manually went through the domestic and cull labelled quality classes and collected twice the number of comparisons as there were images in each class for the training and validation sets. We did not need to collect comparisons for the export labelled class, as all the fruit within it are of identical quality (no defects), and so they are all considered equal. To ensure accurate test data, we performed the test set comparison labelling twice (on the same pairs of cherries both times) roughly one week apart to check for inconsistencies. We also did more comparisons relative to the number of images for the test set than for the training or validation sets. We saw 88.2% and 88.5% agreement between the two rounds of test set comparisons for the cull and domestic sets respectively. To join the two rounds together into one final test set, we checked all the discrepancies and either corrected obvious user mistakes or, in the case of a disagreement without clear error, set the label to equal (as, if neither fruit is clearly better than the other, they must be roughly the same quality). Ultimately, we collected 4074 training (1758 domestic, 2316 cull), 502 validation (216 domestic, 286 cull), and 919 testing (478 domestic, 441 cull) within-class comparisons.

To reduce the impact of noise due to user input error for all sets, we converted the collected comparison information into a directed graph, where nodes represented images, and an edge from node A to B indicated that cherry A was better than B. The graph represented equal labels as bidirectional edges. We

then searched for any cycles in the graph (using the *NetworkX* library's [25] implementation of Johnson's algorithm [26]). We filtered out any cycles consisting of only equal comparisons, and so the remaining cycles indicated an inconsistent set of comparisons (e.g., A $\rightarrow$ B, B $=$ C, C $\rightarrow$ A). Going through any such cycles usually uncovered clear input errors.

B. Network

The network architecture is outlined in Figure 3. During training, the Siamese network passes in two images (for which there exists a comparison) to the same base model that outputs a single scalar value for each image. The two values are subtracted, and then passed through a sigmoid function to produce a final output representing the probability that the first input is better than the second. For the base model, we modify VGG-16 [20] by adding batch normalisation [27] after each max pooling layer, as well as adjusting for inputs of shape 140×140×4 (rather than the original 224×224×3). For the base model, we use a ReLU activation function for all applicable layers aside from the final output, where we have no activation function. All convolutional layers use 3×3 kernels, and all max pooling layers use 2×2 windows. Both have a stride of one. We use VGG-16 as the foundation for our model as, over the years, it has been regularly used with reliable performance. Additionally, 25% of CNN fruit image processing architectures use a VGG model [17], more than any other pre-determined architecture – indicating its potential within the field.

C. Training

During training, the image pairs were given as input to the Siamese network along with a target label. This label represented the probability of the first input being better than the second. Ground truth labels were one of: 1, 0.5, or 0, respectively corresponding to the first input being: better, roughly equal, or worse than the second input. We use binary cross-entropy as the loss function. We train the model using the Adam optimiser [28], for 300 epochs, with a batch size of 16, and a learning rate of 0.0001. Layer weights are initialised using the He uniform scheme [29]. To help reduce overfitting, we add L^2 regularisation (with $\lambda = 0.01$) to the two non-output fully connected layers of the base model. The biases for these two layers are initialised as zeros. We also employ early stopping, ultimately using the iteration of the model that achieved the highest validation accuracy during training.

Additionally, we used the *imgaug* library [30] to apply the augmentation pipeline outlined in Table I, as it allowed for both combined and separate augmentations of the RGB and NIR channels. Images were augmented each time they were encountered during training, meaning the exact same input was never seen by the model more than once.

A mix of between-class and within-class comparisons were used (where "class" refers to the three labelled quality grades), with the latter being the pairwise comparisons discussed in subsection III-A. Each epoch, all the collected within-class labels (for the domestic and cull classes) were used. We tested with and without training using the equal within-class comparisons (assigned a 0.5 label) and found that including them improved performance (when trained for the same number of total comparisons). We also tried generating within-class comparisons for the export class by assigning randomly selected pairs of fruit a 0.5 label (they are all considered to be of equal quality), but this was detrimental. For the between-class comparisons, we randomly paired up each image in the training set with another image (always from a different class) and assigned labels based on which of the pair was from the better quality class. Each epoch, the order of comparisons was shuffled, and all the between-class labels were re-generated.

TABLE I
THE AUGMENTATION PIPELINE. SOME AUGMENTATIONS ONLY OCCUR WITH A CERTAIN PROBABILITY. THE VALUES USED ARE RANDOMLY SAMPLED FROM THE INDICATED RANGE.

Augmentation	Probability	Value Range
Pixel-wise multiply (brightness)	1.0	(0.5, 1.5)
Horizontal flip	0.5	-
Vertical flip	0.5	-
Horizontal scale	1.0	(0.8, 1.2) - proportion
Vertical scale	1.0	(0.8, 1.2) - proportion
Horizontal translation	1.0	(-0.1, 0.1) - proportion
Vertical translation	1.0	(-0.1, 0.1) - proportion
Shear	1.0	(-8, 8) - degrees
Rotation	1.0	(0, 360) - degrees
Gaussian blur	0.25	(0.0, 3.0) - sigma

D. Evaluation

For evaluation, we use three metrics: pairwise within-class accuracy, pairwise between-class accuracy, and three-class accuracy. The first two check whether the Siamese model predicts the correct pairwise comparison for two given images. In contrast to during training, we do not include any equal (0.5) labels here. This is because, unlike for better/worse comparisons which are simply concerned with which input has a higher predicted quality, we can only define an equal prediction in terms of a threshold on how close together the predicted qualities are. Such a threshold would be subjective, and so would make objective evaluation of the model difficult. Additionally, there are relatively few equal comparisons (about a tenth of the overall test set). As such, although we use equal comparisons during training (they still provide additional information), for evaluation we are only concerned with how accurately the model can identify which of two given fruit is of better quality (between and within classes).

The three-class accuracy metric examines how well the fruit can be separated back into the three original classes using the base model's predictions. To do so, we need to find optimal class thresholds in terms of the predicted quality values. We fit a decision tree on these predictions for the training set images (with their class labels as the target) to find these thresholds, and evaluate this fit on the testing data. See Figure 5 for the result of this. Overall, we are primarily concerned with within-class accuracy, as it informs us about how well the model is able to rank fruit within a class, which is precisely what we are interested in.

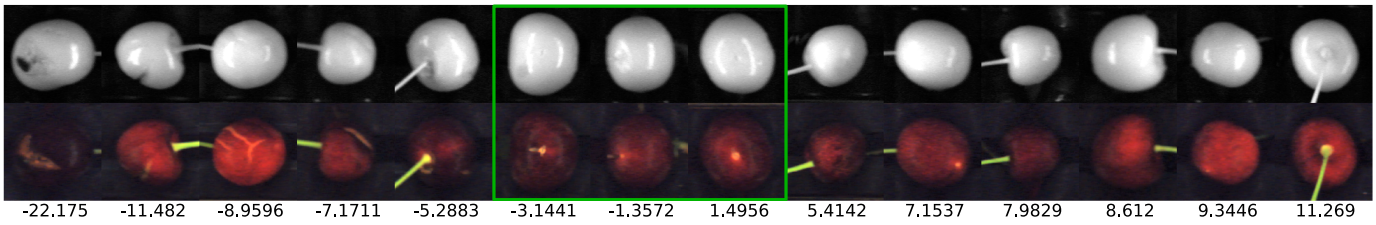


Fig. 4. Every 38th image of the test set as ordered by the base model (of the Siamese network). Values shown are the predicted qualities. As the model learns entirely from comparisons, there is no absolute interpretation of the quality values. Rather, the significance comes from the relative differences between fruit.

TABLE II
EVALUATION OF OUR FINAL MODEL AGAINST HUMAN PERFORMANCE AND A STANDARD CLASSIFICATION MODEL.

	Within-class pairwise accuracy	Between-class pairwise accuracy	Three-class accuracy
Human self-agreement	93.1%	-	-
Standard classifier	60.4%	98.4%	92.6%
Siamese network	88.1%	98.0%	87.9%

IV. RESULTS

We compare our model's performance (on the held-out test set) to the self-agreement of a human expert, as well as to a standard classification model. We previously stated an 88% human agreement for within-class comparisons, which included the disagreements where one instance labelled a comparison as equal while the other did not. As we cannot objectively use the equal label when evaluating the model (as discussed in subsection III-D), for a fair comparison we only consider better/worse labels for the human self-agreement as well. Doing so, we find an agreement of 93.1%.

The standard classification model consists of the same architecture as our base model, except its output is a probability distribution over the three quality classes. To use this for within-class comparisons, we took the expected value of the predicted distribution as the quality score. The results are outlined in Table II. We see that, although not quite at human performance, our model does well on within-class comparisons (just 5.0% below human), far out performing the standard classification model. Additionally, remember that the model's overall accuracy will be higher than the within-class accuracy suggests, as many comparisons will be between classes. While our approach is able to achieve almost the same accuracy as the standard classification model for between-class comparisons, one downside is that it performs 4.7% worse in terms of three-class accuracy; however, this is to be expected, as we are intentionally trying to blur the lines between classes.

We examine the distribution of the predicted quality values in Figure 5. There is little overlap between the export and domestic classes, while there is notably more between domestic and cull. This is not surprising, as there is a clearer distinction between export and domestic (no defect versus some defect) than between domestic and cull (moderate defect versus severe defect). Additionally, it displays a major outlier in the domestic class, which we investigated further. The outlier had no surface defects, but was labelled as deformed due to its peculiar shape. There were fewer than five such fruit in the training set (indicating there is still potential for improvement by collecting/utilising more data), so it is hardly surprising the model was unable to deal with it correctly.

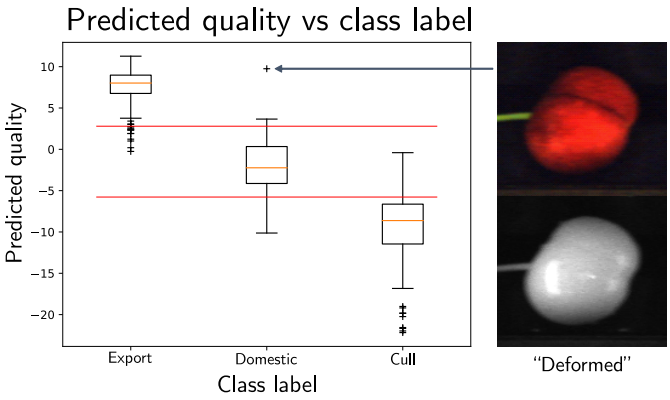


Fig. 5. The distribution of quality labels predicted by the base model (extracted from the Siamese network) on the test set, plotted against the class labels. Red lines indicate the quality thresholds extracted (using a decision tree) from the training set. The cherry displayed (on the right) corresponds to the indicated outlier.

To illustrate the overall ordering performed by the model, we have it rank all the fruit in the test set from worst to best. Figure 4 displays this, plotting every 38th image. We see that the model indeed appears to be ordering the fruit, giving a cherry with a rotten crack the lowest quality score and perfect cherries the highest ones. Furthermore, the fruit smoothly blend from worst to best. We also note two other promising occurrences: the three cherries highlighted in green display small nose cracks that decrease in severity, which the predicted quality scores reflect perfectly; and the low value given to the second image (from the left) shows the model correctly recognising that the crack is especially severe (which we can see from the NIR image).

V. CONCLUSION

We have seen that, while the grading of cherries is critical for the industry, the technology backing it has fallen behind the current computer vision trends. One of the major issues in

incorporating DL into the grading process is that conventional classification methods are not able to handle an environment where the class definitions themselves regularly change. To account for the shifting grade standards, we transform the problem from classification to regression, using a Siamese network to train a model that can predict continuous quality values while only learning from pairwise comparisons. We find that, although not yet performing at human accuracy, our model comes close. However, in blurring the lines between the original classes, it also becomes slightly worse at distinguishing between them. Inference can be performed at speeds suitable for deployment (mean processing time is increased by only 1.25ms/fruit on commodity hardware), and the model will be trialled in production during the 2020-2021 season.

In the future, performance could be improved by incorporating large amounts of unlabelled data into the training process – using semi-supervised learning – as collecting raw data is cheap, while labelling it is very time consuming. Additionally, the labelling of pairwise comparisons could be made more effective by only presenting the human with instances the model is unsure about. Overall, we have found a very promising solution to a major roadblock in the cherry grading process. As New Zealand must compete with larger cherry exporting countries (such as Chile and Canada) that can process far larger quantities of fruit, we place more importance on quality than quantity. By developing a model able to make fine-grained quality decisions, we hope to provide New Zealand's industry with a powerful, competitive new tool.

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